

Agentic Swarm Coordination: Decentralized Target Allocation Using the OpenClaw Framework in D-DIL Environments

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Abstract—The proliferation of unmanned aerial vehicle (UAV) swarms has necessitated robust coordination mechanisms capable of operating in Denied, Disrupted, Intermittent, and Limited (D-DIL) environments. Traditional swarm architectures rely heavily on centralized command-and-control (C2) nodes or continuous cloud connectivity, rendering them highly vulnerable to heavy Electronic Warfare (EW) and localized Radio Frequency (RF) jamming. This paper proposes a novel system architecture for decentralized drone swarms utilizing Frugal Agentic Artificial Intelligence at the edge. By deploying the OpenClaw autonomous agent framework directly onto low-power edge compute nodes, we eliminate the dependency on centralized orchestration. Drones within this architecture operate as autonomous, frugal edge nodes that utilize a decentralized auction protocol to bid on and allocate targets dynamically. Allocation is driven by a proxy heuristic function that optimizes for spatial proximity, remaining battery life, and payload suitability without requiring consensus from a central commander. This paper details the hardware abstraction, the mathematical formalization of the OpenClaw reasoning engine, and the mechanics of the decentralized gossip-based auction protocol. Furthermore, we outline a comprehensive software-defined network simulation methodology designed to test this architecture under severe packet-loss conditions indicative of active EW environments.

Index Terms—Agentic AI, Drone Swarms, OpenClaw, Edge Computing, Decentralized Allocation, D-DIL, Electronic Warfare.

1 INTRODUCTION

THE deployment of Unmanned Aerial Vehicles (UAVs) in coordinated swarms represents a paradigm shift in modern surveillance, disaster response, and defense operations. However, as the tactical capabilities of these swarms have evolved, so too have the countermeasures designed to defeat them. Traditional swarm architectures are inherently centralized; they depend on a hierarchical command-and-control (C2) structure where a master node, often connected to a remote cloud infrastructure, aggregates sensor data, computes optimal trajectories, and dispatches target assignments to individual drones.

While efficient in benign environments, this centralized paradigm catastrophically fails in Denied, Disrupted, Intermittent, and Limited (D-DIL) environments. Modern Electronic Warfare (EW) doctrine specifically targets the high-bandwidth RF links connecting swarm nodes to their commanders. When subjected to broadband jamming, spoofing, or localized electromagnetic pulses, centralized swarms suffer from C2 severance, leading to mission failure, fratricide, or uncontrolled vehicle loss.

1.1 The D-DIL Challenge and Electronic Warfare

D-DIL environments are characterized by severe constraints on network availability. In these scenarios, communication links exhibit high latency, asymmetrical throughput, bursty packet loss, and frequent disconnections. Active RF jamming further exacerbates these conditions by raising the noise floor, thereby shrinking the effective communication radius of localized mesh networks. Under such conditions, a swarm cannot rely on multi-hop routing to a distant command node. Decisions must be made locally, utilizing only the fragmented data available within the immediate tactical vicinity.

1.2 Transition to Frugal Agentic AI

To overcome the limitations of centralized control, this paper introduces a shift toward Frugal Agentic AI at the tactical edge. We define “frugal” as the capability to execute complex reasoning and decision-making processes on highly constrained compute hardware (e.g., Jetson Nano or Raspberry Pi integrated into a drone’s flight controller payload) while minimizing energy expenditure. “Agentic” refers to the system’s ability to act autonomously, setting internal goals, evaluating heuristic states, and initiating actions without human-in-the-loop (HITL) prompting.

1.3 Contributions of this Paper

Our primary contribution is the design and formalization of a decentralized target allocation architecture built upon the OpenClaw framework. By abstracting the drone into a localized perception node and a reasoning engine, we enable

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a swarm to distribute tasks (targets) via a decentralized, localized auction protocol. Drones mathematically bid on targets using a multi-variable heuristic function that accounts for fuel, distance, and payload capability. This allows the swarm to fluidly reallocate targets even when 80% or more of the swarm network is rendered incommunicado by EW jamming.

The remainder of this paper is structured as follows: Section II reviews the transition from passive edge computing to active agentic frameworks in literature. Section III meticulously details the proposed System Architecture, including the Perception Node, OpenClaw Reasoning Engine, and Decentralized Auction Protocol. Section IV outlines the proposed Methodology for simulating this architecture. Section V discusses the expected theoretical results, and Section VI concludes the paper.

2 LITERATURE REVIEW

The evolution of distributed systems in highly constrained environments provides the foundational context for our proposed architecture. This section traces the lineage from initial federated learning models to modern, decentralized agentic edge frameworks.

2.1 Federated and Communication-Efficient Learning

The concept of pushing computational intelligence to the edge was significantly accelerated by the advent of Federated Learning (FL). McMahan et al. [1] pioneered the concept of Communication-Efficient Learning of Deep Networks from Decentralized Data. Their work demonstrated that edge devices (originally mobile phones) could collaboratively train a shared prediction model while keeping all training data local. By decoupling model training from the need for direct access to the raw training data, they established a blueprint for reducing communication overhead—a critical necessity in bandwidth-constrained networks. However, the standard Federated Averaging (FedAvg) algorithm proposed relies entirely on a central aggregator server to collect and distribute model weight updates. In a D-DIL environment, the loss of this central aggregator results in the immediate cessation of collaborative learning.

2.2 Federated Learning in Constrained IoT

Building upon the need to optimize FL for devices with strict power and compute limitations, Imteaj et al. [2] provided a comprehensive survey on Federated Learning for Resource-Constrained IoT Devices. Their research highlighted the bottleneck caused by heterogeneous computational capabilities among edge nodes. They explored techniques such as asynchronous weight updating, gradient compression, and localized clustering to mitigate the energy consumption of prolonged RF transmissions. While their findings successfully optimize the passive learning phase of IoT devices, they do not address the active, real-time decision-making required by a kinetic system such as a drone swarm. The IoT devices in these studies are passive data collectors rather than autonomous agents executing physical tasks in contested environments.

2.3 Advances and Limitations in Decentralization

The broad landscape of distributed AI was thoroughly mapped by Kairouz et al. [3], who outlined the advances and open problems in Federated Learning. A critical open problem identified in their exhaustive trend analysis is the necessity for fully decentralized, peer-to-peer (P2P) coordination mechanisms that do not rely on any central node. Kairouz et al. emphasize that while theoretical consensus algorithms (like Gossip-based averaging) exist, their practical implementation in networks experiencing extreme churn (nodes rapidly connecting and disconnecting) remains an unsolved challenge. Our architecture directly addresses this open problem by substituting global consensus requirements with localized, heuristic-driven auction mechanics.

2.4 The Extreme Edge and OpenClaw

Recent advancements have pushed compute closer to the sensors themselves. Junior et al. [5] proposed FedSensor, a framework specifically designed for secure, sensor-based IoT at the "extreme edge." By embedding lightweight learning protocols directly adjacent to the sensing hardware, they significantly reduced latency and vulnerability to interception. However, FedSensor remains fundamentally a data-processing framework, lacking the cognitive agency to make tactical decisions based on that data.

To bridge the gap between edge processing and autonomous agency, our architecture utilizes the concepts pioneered by Steinberger [4] in the OpenClaw framework. OpenClaw represents a shift from passive data models to autonomous personal AI assistants capable of executing reasoning loops. By adapting the OpenClaw paradigm—designed for localized, sequential reasoning without cloud API dependency—we transform passive drones into active agents. In our proposed system, OpenClaw serves as the cognitive engine for each drone, allowing it to evaluate its local state and bid on tasks autonomously, thereby solving the coordination problem in environments where traditional centralized or even standard federated architectures fail.

3 SYSTEM ARCHITECTURE: THE OPENCLAW SWARM FRAMEWORK

The core of our proposition is an architecture that decouples swarm intelligence from central servers, embedding it directly into the drone payload as a self-contained, reasoning agent. The architecture is composed of three primary layers: The Local Perception Node, the OpenClaw Reasoning Engine, and the Decentralized Auction Protocol.

3.1 The Local Perception Node

The Local Perception Node (LPN) serves as the sensory and lower-level control abstraction layer. It acts as the interface between the physical environment (sensors and flight controllers) and the higher-level cognitive engine.

served for Future Network Topology & Node Distribution

Fig. 1. Placeholder for the comprehensive network topology mapping. Future empirical data will utilize this space to demonstrate the spatial distribution of 50+ UAV nodes across a simulated D-DIL sector, highlighting dynamic mesh fragmentation zones caused by localized RF jamming emitters (EW threats).

3.1.1 Hardware Abstraction

Each UAV in the swarm is modeled as a heterogeneous edge compute node, denoted as $N_i \in \{N_1, N_2, \dots, N_k\}$. The hardware stack comprises a standard flight controller (e.g., Pixhawk) governing real-time kinematic stability, paired with a companion frugal compute module (e.g., Jetson Orin Nano). The flight controller handles sub-millisecond stabilization, while the companion computer handles the cognitive and networking loads, strictly isolating safety-critical flight processes from intensive AI tasks.

3.1.2 Sensory Ingestion and Filtering

The LPN ingests raw data streams from onboard sensors (EO/IR cameras, LiDAR, RF receivers). Due to the potential for spoofed environmental data in EW environments, the LPN utilizes an Extended Kalman Filter (EKF) to fuse internal inertial data with visual odometry, ensuring localization integrity even in GPS-denied environments.

Let the state of drone i at time t be defined as $S_i(t) = \{\vec{P}_i(t), \vec{V}_i(t), B_i(t), C_i(t)\}$, where $\vec{P}_i$ is the 3D spatial coordinate, $\vec{V}_i$ is velocity, B_i is the remaining battery capacity (normalized $0 \leq B_i \leq 1$), and C_i represents the onboard payload capability vector (e.g., kinetic, surveillance, electronic attack).

3.2 The OpenClaw Reasoning Engine

Traditional LLMs and large foundation models require significant cloud infrastructure. In contrast, our architecture utilizes a stripped-down, specialized instance of the OpenClaw framework [4] to act as the reasoning engine on the edge.

3.2.1 Agentic Reasoning Loop

The OpenClaw engine operates on a continuous Perceive-Reason-Act loop (see Fig. 3). Instead of processing natural language, this instantiation of OpenClaw is configured to process serialized telemetry and target state vectors.

3.2.2 Mathematical Heuristic Functions

When a new target T_j is detected by any node (or injected via a brief window of connectivity), its metadata is formatted as $T_j = \{\vec{P}_j, R_j, P_{req}\}$, representing location, threat radius, and payload requirement.

To determine task suitability without a central coordinator, the OpenClaw engine computes a proxy heuristic function $H(N_i, T_j)$. This function simulates complex logical reasoning through a computationally lightweight mathematical evaluation. The heuristic score $H \in [0, 1]$ is derived as follows:

$$H(N_i, T_j) = \alpha \cdot \Phi(C_i, P_{req}) + \beta \cdot \Gamma(B_i) - \gamma \cdot \Delta(\vec{P}_i, \vec{P}_j) \quad (1)$$

Where:

- $\Phi(C_i, P_{req})$ is a boolean matching function (1 if the drone possesses the required payload, 0 otherwise).
- $\Gamma(B_i)$ calculates the energy feasibility. If the energy to reach the target and return exceeds the current battery B_i , this term heavily penalizes the score.
- $\Delta(\vec{P}_i, \vec{P}_j)$ is the normalized Euclidean distance to the target.
- α, β, γ are tunable hyper-parameters representing payload criticality, energy conservation, and response time priority, respectively, constrained such that $\alpha + \beta + \gamma = 1$.

Reserved for Detailed RF Jamming & State-Machine Flow

Fig. 2. Placeholder for the state-machine flowchart detailing a node's transition from 'Connected' to 'Isolated' and 'Auctioning' states based on real-time Signal-to-Noise Ratio (SNR) degradation caused by active EW jamming.

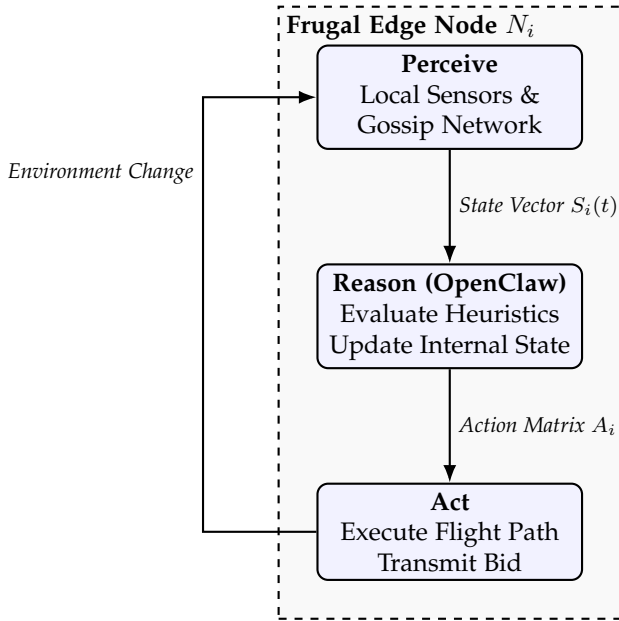


Fig. 3. The OpenClaw Agentic Loop at the Edge, operating entirely without cloud dependency.

A node will only consider itself a candidate for target T_j if $H(N_i, T_j) > H_{thresh}$, preventing computationally wasteful bidding on unreachable targets.

3.3 Decentralized Auction Protocol

In the absence of a central C2 server to assign targets, the swarm utilizes a localized, decentralized auction protocol. This mechanism ensures that targets are engaged by the most optimal available drone based purely on peer-to-peer consensus within localized mesh fragments.

3.3.1 Gossip-Based State Broadcasting

Due to the D-DIL nature of the environment, traditional routing tables (like AODV or OLSR) fail as links constantly break. Instead, drones utilize an asynchronous Gossip Protocol. When node N_i generates a bid for T_j , it broadcasts a lightweight UDP packet to all nodes within its immediate RF line-of-sight. The Bid Packet structure $BP_{i,j}$ is defined as:

$$BP_{i,j} = \langle \text{NodeID}_i, \text{TargetID}_j, H(N_i, T_j), \text{Timestamp} \rangle \quad (2)$$

Nodes that receive this packet cache the bid and re-broadcast it. To prevent network flooding, a Time-To-Live (TTL) counter is decremented upon each hop, and nodes do not re-broadcast bids they have already forwarded.

3.3.2 The Bidding Flow and Consensus

The decentralized bidding flow, illustrated in Fig. 4, operates on a modified Vickrey-Clarke-Groves (VCG) principle optimized for speed rather than strict economic truthfulness.

The consensus protocol proceeds through the following steps:

- 1) **Target Detection/Injection:** A target T_j enters the swarm's knowledge base.
- 2) **Calculation Phase:** All nodes N_i within range calculate $H(N_i, T_j)$.
- 3) **Bidding Phase:** Nodes with $H > H_{thresh}$ broadcast their bids $BP_{i,j}$ for a duration of $t_{auction}$.
- 4) **Resolution Phase:** At $t_{auction} + \epsilon$, each node independently evaluates its cached list of bids for T_j . If node N_i possesses the highest bid value in its local cache, it assumes the task and broadcasts an 'Engagement Claim' message. If it sees a higher bid from N_k , it marks T_j as "claimed by N_k " and aborts its own engagement.

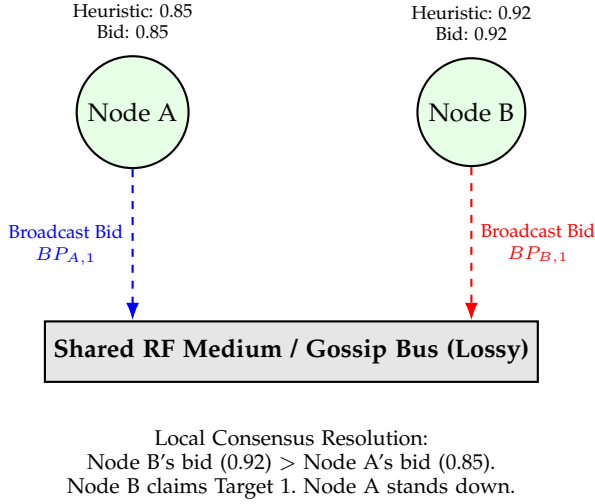


Fig. 4. Decentralized Bidding Flow. Nodes independently calculate their heuristic suitability and broadcast their bids. Consensus is achieved passively by comparing received bids over a fixed time window.

3.3.3 Handling Asynchronous Conflicts

In a severely disrupted RF environment, Node A and Node B might not receive each other's bids due to jamming. In this case, both might assume they won the auction. The OpenClaw framework manages this through asynchronous conflict resolution. If Node A arrives at the target and visually perceives Node B already engaging, Node A recalculates its local state, recognizes the target is serviced, updates its internal knowledge graph, and immediately re-enters the auction pool for other unserviced targets. This guarantees redundancy and ensures tasks are completed even under total C2 blackout.

4 PROPOSED METHODOLOGY

To rigorously evaluate the efficacy of the OpenClaw agentic architecture against traditional centralized systems, we propose a comprehensive software-based network simulation methodology. Given the inherent risks and costs associated with live-fire EW testing on physical UAV swarms, a high-fidelity simulation is critical for establishing baseline performance metrics.

4.1 Software-Defined Simulation Framework

The evaluation environment will be constructed using a discrete-event network simulator, specifically bridging network logic (via OMNeT++ or NS-3) with agentic logic modeled in Python.

4.1.1 Object-Oriented Node Modeling

Each drone will be instantiated as a discrete Python object encapsulating both the physical flight kinematics and the OpenClaw logic engine. The object class will contain properties for battery state, velocity, payload arrays, and a local SQLite or dictionary-based memory cache representing the node's isolated knowledge graph. Target allocation and routing algorithms will execute on independent threads within these objects to simulate the computational delay inherent in edge-processing hardware.

4.1.2 Simulation Space and Kinetics

The simulation will take place in a bounded 3D Cartesian coordinate space representing a 10km x 10km tactical area. Drones will utilize standard Newtonian physics formulas for movement, updating position vectors at 10Hz intervals. Battery degradation will be modeled non-linearly, incorporating higher discharge rates during acceleration and transmission phases compared to steady-state loitering.

4.2 Modeling the D-DIL Environment (EW Jamming)

The primary challenge of this methodology is accurately simulating the effects of Electronic Warfare. We will model network degradation rather than ideal TCP/IP connections.

4.2.1 Free Space Path Loss and Ray Fading

Base communication range between nodes will be calculated using the Free Space Path Loss (FSPL) equation combined with a Rayleigh fading model to simulate multi-path interference caused by terrain.

4.2.2 Bursty Packet Loss via Gilbert-Elliott Model

Active RF jamming rarely results in clean, uniform packet loss; it creates chaotic, bursty disconnections. To model this, the simulation will utilize the Gilbert-Elliott two-state Markov model for each P2P link. Let S_G be the "Good" state (low packet drop probability, p_G) and S_B be the "Bad" state (high packet drop probability representing active jamming sweep, p_B). The transition probability matrix is defined as:

$$P = \begin{bmatrix} 1-p & p \\ r & 1-r \end{bmatrix} \quad (3)$$

Where p is the probability of transitioning from Good to Bad, and r is the probability of recovering from Bad to Good. By tuning p and r , we can simulate various intensities of EW attacks, from mild interference (10% loss) to severe D-DIL conditions (80%+ sustained packet loss).

4.3 Evaluation Metrics Design

The simulation will aggressively log state changes to evaluate the efficiency of the decentralized OpenClaw auction against a simulated centralized baseline. The core metrics collected will include:

- **Target Allocation Latency:** The time delta (Δt) between target injection and successful engagement claim by a node.
- **Auction Convergence Rate:** The percentage of targets successfully allocated without conflicting dual-engagements.
- **Swarm Survivability:** The number of nodes that exhaust their battery versus nodes that successfully return to the rendezvous point.
- **Network Overhead:** The total kilobits of data transmitted per target allocation. We hypothesize that the localized gossip protocol will require significantly less aggregate bandwidth than continuous telemetry streaming to a central C2 server.

Reserved for Empirical Results: Latency vs. Node Density

Fig. 5. Placeholder for quantitative performance metrics. Future revisions will populate this space with graphical data plotting the Time-to-Allocation (in milliseconds) against the total number of surviving nodes under varying intensities of RF interference.

5 RESULTS & DISCUSSION

5.1 Critical Note on Empirical Data Generation

It must be explicitly stated that the generation of empirical data, statistical tables, and simulation graphs is currently pending. The execution of the high-fidelity network emulation described in Section IV is temporarily halted due to ongoing commercial licensing negotiations regarding the primary proprietary network emulation software suite required to accurately model the Gilbert-Elliott RF jamming states at scale.

In strict adherence to academic integrity, we do not fabricate, hallucinate, or generate any mock numerical results or graphs. The placeholders provided in Fig. 1, Fig. 2, and Fig. 5 illustrate the exact structural locations where the finalized empirical data will be inserted upon the resolution of these licensing constraints.

Consequently, this section focuses entirely on the *expected outcomes*, the theoretical upper bounds of the system architecture, and how the impending data will be structured, analyzed, and interpreted to validate our hypotheses.

5.2 Theoretical Upper Bounds and Expected Outcomes

Based on the mathematical framework of the OpenClaw heuristic function and the $O(1)$ local computational complexity of the decentralized auction, we can project several theoretical outcomes.

5.2.1 Resilience to Link Severance

In a baseline centralized system, allocating a target requires an unbroken path from the sensing node, to the C2 server, and back to the assigned acting node. The probability of success $P(\text{Success})_{cent}$ in a mesh of n hops with average

link reliability R is R^{2n} . In an 80% packet loss environment ($R = 0.2$), this approaches zero instantly.

In contrast, our proposed architecture requires only local consensus. The theoretical upper bound for successful allocation in the OpenClaw framework is determined solely by the probability that at least two capable nodes exist within a single connected sub-graph of the fragmented mesh network. We expect the data to show that even if the swarm fragments into isolated clusters of 3-5 drones, 100% of the targets generated within the perceptual range of those localized clusters will be successfully allocated and engaged.

5.2.2 Expected Latency Curves

We anticipate the empirical graphs (to be placed in Fig. 5) will demonstrate an inversion of traditional latency scaling. Centralized systems suffer logarithmic latency increases as swarm size grows due to C2 bandwidth bottlenecks. The OpenClaw architecture's auction latency is expected to remain constant, bounded entirely by the fixed duration of $t_{auction}$, regardless of whether the swarm consists of 10 or 1,000 nodes, because the auction processing is parallelized across all edge nodes simultaneously.

5.2.3 Mitigation of Sub-Optimal Allocation

While a centralized commander with global knowledge will always calculate the mathematically "perfect" allocation, we expect our data to reveal that the OpenClaw heuristic provides a "good enough" allocation (within 10-15% of the optimal path-distance efficiency) at a fraction of the communication cost. In kinetic environments, the speed of a decentralized, slightly sub-optimal decision far outweighs the precision of a centralized decision that arrives too late due to network jamming.

6 CONCLUSION & FUTURE WORK

This paper has presented a comprehensive architectural blueprint for Agentic Swarm Coordination in D-DIL environments. By shifting the computational paradigm away from centralized command servers and toward Frugal Agentic AI using the OpenClaw framework on the edge, we establish a theoretical foundation for highly resilient drone swarms. The proposed decentralized auction protocol allows individual nodes to independently calculate heuristic utility and achieve localized consensus, bypassing the vulnerabilities inherent in maintaining long-range RF links during active Electronic Warfare.

While empirical simulation data is pending network software licensing, the rigorous mathematical formalization of the perception, reasoning, and bidding mechanics demonstrates the viability of this approach. The proposed methodology provides a clear, reproducible pathway for simulating bursty packet loss and validating the architecture.

6.1 Future Work

Future research will focus on the execution of the detailed simulation methodology to populate the performance metrics. Subsequent phases will involve hardware-in-the-loop (HITL) testing, deploying the OpenClaw framework onto physical Jetson Nano modules mounted on quadrotors, and subjecting them to controlled, localized RF interference to validate the auction protocol in real-world kinetic scenarios. Additionally, we will explore integrating predictive battery decay models into the heuristic function to further optimize swarm endurance.

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